

USING IMPACT TO MAKE IMPACT?

Experiences from a dialect crowdsourcing project

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Introduction

The English Dialects App, a smartphone application aimed at both engaging the English public about language variation and crowdsourcing present-day dialect data, was released in January 2016.¹ It was the result of the collaboration of a group of Swiss phoneticians, app developers and MA students with an English dialectologist. Adrian, Marie-José and their team had previously developed a highly successful and impactful app for Swiss German dialects, the *Dialäkt Äpp*, and asked Dave to work with them to develop one suitable for England. The aims of the English Dialects App were (at least) three-fold: partly as a fun and accessible way to engage the public about dialect and demonstrate that language variation is contemporary and alive; partly to gather, in a manner that was economical with both time and money, a snapshot of the geographical distribution of language variation that was also sensitive to the social backgrounds of the dialect users – the last nationwide survey of English dialects dates back to the 1950s and 1960s when interviewees were overwhelmingly over 60 years old, male, rural and non-mobile; and partly also to trouble and critique our own complacencies about dialectological fieldwork methods and about ‘authentic’ dialect data.

In order for the app to fulfil these aims, we needed to make sure people knew it existed – we needed visibility. Before the release, many hours were spent thinking about effective dissemination methods, and talking to the media and our university press offices. At its release it did make an impact, in the sense of considerable media coverage. The app secured lots of users in the first few months, so much so that we were able to update the functions of the app, re-release in May 2016 and secure more news attention because we were able to present results from our analysis of the crowdsourced data. Users continue to download the app to this day, and occasionally the media re-find the results from 2016 and re-tell parts of the story. In the sense of engagement with the general public, therefore, the English Dialects App made impact. We therefore achieved the first of our above-mentioned aims, as the app succeeded in connecting with a high number of users who learned about and interacted with accent and dialect-themed questions. We argue below, however, that the app data potentially have further very important, but less immediately newsworthy applications.

In this chapter, we reflect upon the process of using ‘impact’ to make ‘impact’ (academic and beyond), the longer term benefits of the app outside of academic linguistics and our experiences of the impact agenda from our own different institutional trajectories – someone (Dave Britain) who moved from the UK to Switzerland (where the impact agenda is in its early infancy), someone (Adrian Leemann) who moved from Switzerland to the UK and someone (Marie-José Kolly) who moved from academia to data journalism. We begin, however, by briefly introducing the app itself, since it was explicitly designed to gain exposure, uptake and impact.

The English Dialects App (EDA)

EDA is a free iOS and Android app with two core functionalities – a dialect quiz and dialect recordings. For the quiz, users indicate which variants of 26 words they themselves use and the application attempts to geo-locate their dialect; users can also upload recordings of themselves reading a short version of the ‘Boy who Cried Wolf’ story and they can listen to recordings of those submitted by other users, as well as recordings of speakers from the Survey of English Dialects (Orton et al. 1962–71) from the mid-twentieth century. The app had by May 2017 been downloaded more than 99,000 times. More than 50,000 users from more than 4,900 localities across the UK have chosen dialect variants for the 26 words, and more than 4,300 users submitted

audio recordings. The app's functions incorporated elements from an earlier app that had been developed for Swiss German (Leemann and Kolly 2013; Kolly and Leemann 2015; Leemann et al. 2015), the *Dialäkt Äpp*, thus both saving costs and building on earlier very successful work.

As for the English Dialects App's second aim, the function of the quiz was to gather data on lexical, phonological and grammatical variables. Each screen in the quiz prompts users to either answer a question or complete a fill-the-gap task (see Figure 7.1). The possible answers are listed below the question. For the phonological variables, users can listen to the possible answers by clicking on the circled arrow to the left of the answer. They would then be moved to the next screen, which would show a map of where in the country their answer was also used and asked the next question. Typical phonological questions can be seen in Figure 7.1. We also asked some lexical questions, such as 'What do you call a piece of wood stuck under the skin', morphological questions, such as 'It belongs to the woman, not the man. It's not his, it's ___' and 'Every day on her walk past the lake, she ____ the ducks: do feed, feed, feeds', as well as syntactic ones, such as 'If you'd like someone to pass you something, would you say: 'give it me', 'give me it' or 'give it to me?'. Different questions have different numbers of possible answers, ranging from two to ten (for the piece of wood question). Once all 26 quiz questions have been answered, the app predicts where that user was probably from. The basis for the prediction is 25 maps of the same linguistic variables from the Survey of English Dialects (Orton et al. 1962–71) (SED). The app compares users' quiz answers with the answers given for the same variables by the 313 interviewees from the SED, and presents three locations from the SED that best correspond to the answers given by the users (see the right screen in Figure 7.1).



FIGURE 7.1 Prompts for variant selection (left and centre) and dialect quiz result (right).

If users believe the result to be accurate, they are shown a screen informing them about how to support our research (Figure 7.2, left). By clicking on 'OK, I'm in!', users are asked to indicate their dialect location and complete a set of social background questions on age, gender, ethnicity, mobility and education. Our privacy and ethics policy is explained on the screen, and users are asked to click if they accept. If users feel the first locality presented is not accurate (and as we explain below, we expected the vast majority of people to fall into this category), they can locate their dialect by moving a pin on a map to the place that best corresponds to their dialect (Figure 7.2, centre). In both instances—correct or false dialect prediction—the location of a speaker's actual dialect is therefore elicited. Users also have the opportunity to decline further input into our research, in which case they are shown the results screen again (Figure 7.1, right). Users' answers together with the social background information are then anonymously stored on a server.



FIGURE 7.2 Prompts for evaluating the dialect guessing result: consent form (left), placement of pin to locality that best corresponds to one's dialect (centre) and prompt for metadata (right).

The second core function allows users to anonymously record, listen to and upload their reading of a short passage, listen to old recordings from the Survey of English Dialects and listen to the recordings uploaded by other app users by navigating an interactive map. The aim here was to provide users with the possibility of listening to dialect samples from all over England and thereby build an awareness of dialectal variation, as well as to crowdsource continuous speech of high recording quality, with little to no degradation of the sound quality – possible with today's smartphones. The text we chose was the 'Boy who Cried Wolf' fable, since it contains all the sounds of English, and, for the more variable ones, with sufficient frequency (Deterding 2006).

Here, the app prompts speakers first to self-declare their dialect, again by placing the pin on the locality that best corresponds to their dialect and to fill in the social metadata described above. On the metadata screen, it is explained that in recording their voices, the users consent to our privacy statement. Users can also opt out at any stage. They then proceed to the recording instructions and then read aloud the sentences shown on the screen. Once their recording has been uploaded, users can, by navigating an interactive map, listen to their own recordings and those of others, as well as the historical recordings from the Survey of English Dialects. The info tab on the app provides some information about the app's functions, its methods, its developers and its privacy policy.

We are, as sociolinguists and sociophoneticians, fully aware of the problems and drawbacks of collecting dialect data in this way. The sample gathered with the app is, understandably, biased towards digital natives – it is younger and more educated than the population as a whole. But the collection of metadata from speakers means we can nevertheless compare users of different ages, educational levels, mobilities and so on. We are also aware that the recording function records users in a rather formal style – reading an unfamiliar passage – and the quiz is not collecting dialect use, but, ultimately, intuitions about dialect use. We present an honest account of these factors as well as more detail about the app itself elsewhere (Leemann et al., under review; Britain et al., in preparation).

The data crowdsourced from the app has enabled us to conduct a range of different kinds of analyses. Thus far, most analysis has been conducted on the quiz data. The combination of users' location as well as social background information has enabled us to, for example, produce maps not only showing patterns of variation right across England (a strength of the SED, a weakness of many contemporary social dialect studies of individual communities), but also consider how these patterns differ according to, for example, the age, gender, educational level and mobility of the users (a strength of locally oriented social dialect studies, but a weakness of the SED). We have been able to update our understanding of the geographical position of the 'North-South' dialect divide (Grossenbacher et al. 2017), considering the pronunciation of the vowel in 'butter' – Southern /ʌ/ versus Northern /ʊ/ and in 'last' – Southern /ɑ:/ versus Northern /a/. We've also been able to track the geographical spread of innovations, such as [f] for /θ/ in 'think', as well as those

features which appear to be undergoing rapid decline, such as rhoticity (the pronunciation of non-pre-vocalic /r/ in words such as 'farm' and 'cart'). Figure 7.3 is an example from the app analysis. The variable under scrutiny in these maps is dative alternation: *give it to me*, *give me it* and *give it me*. It shows on the left, the distribution of the three variants in England in the Survey of English Dialects. On the right, the pattern for *give it me* in our app results. This variant appears to have shrunk back from its rather widespread distribution at the time of the SED to a 'heartland' of the Northern Midlands, South Yorkshire and Lancashire. It appears to be robust in this heartland, however. Despite its apparent disappearance from the South, levels of *give it me* use in this heartland are high, given that the SED data are solely from older rural men, whereas the app data are from a wide cross-section of the population, including the young, the urban and the highly mobile.

We move now, though, to considering the development and outcomes of the app in terms of impact – how it shaped the way we designed the app in the first place, what happened after the release, set within the context of us working in rather different academic impact cultures.

Impact 1: preparing for the launch

Achieving the academic goals of the project relied entirely on us being able to achieve the engagement ones. If people didn't know about the app, they wouldn't use its functions and we wouldn't collect any data. We therefore needed (i) a user-oriented design and (ii) a media attention strategy. 'Audience design' therefore clearly had to be central to how we built the app. We therefore had the task of making it easy and attractive to use, entertaining and not too burdensome on the one hand, while gathering the data, for example on phonological variation, that we wanted, on the other. We ensured the app was released for Android as well as iPhones. The Dialäkt Äpp in Switzerland was only available on iPhone, but we were aware that iPhone has a smaller share of the smartphone market in England than it does in Switzerland. Our decision-making in designing the app's functions had to be centred on user experience and, as a consequence, we sometimes felt obliged to prioritise usability over our theoretical research goals. In choosing questions, for example, we tended to shy away from variables with too many different variants, especially phonological ones. We had to strike the right balance with the number of questions in the quiz and how they were asked in order not to overburden users or lead to their boredom. And we had to think about how we could 'create a story' for the app that would engage users and the media (who would, we hoped, help us find those users). The 'dialect prediction function' was the result of this search for an appealing handle for the app – frustrating was the fact that we knew from the start that the prediction function would work poorly because the only baseline we could work from was the SED, data from people born over 125 years before.

Partly also to satisfy the engagement function, but also out of sheer dialectological curiosity, we added one variable which was not in the SED, the pronunciation of the vowel in 'scone'. This was included because the variant pronunciations of it are, for English, highly variable, extremely salient and ideologically heavily loaded – do a search for 'scone' and 'pronunciation' in, for example, Twitter, and you are bombarded with hundreds and hundreds of hits. We expected the inclusion of this variable, therefore, to help spark media and user interest too.

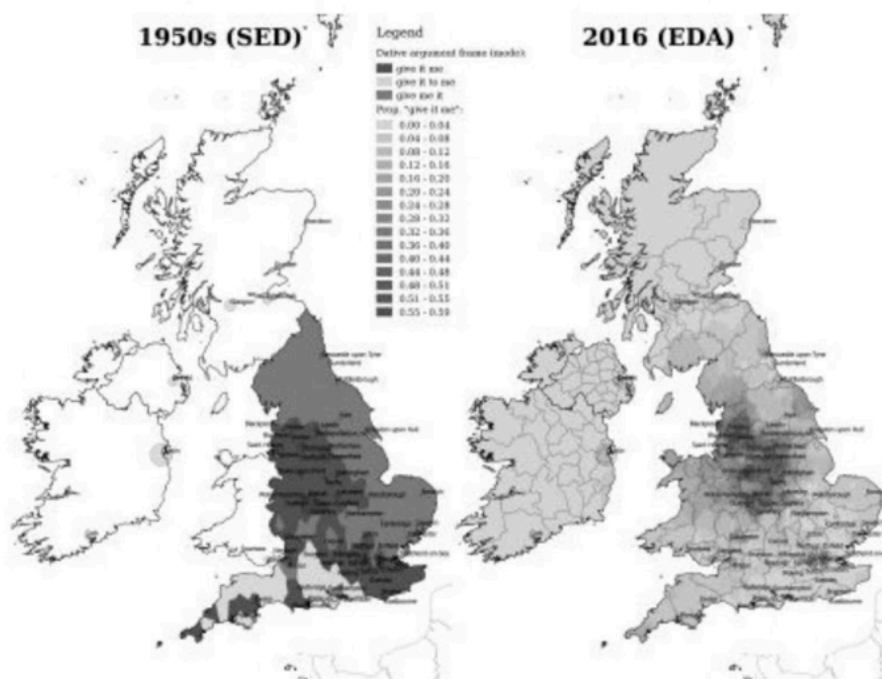


FIGURE 7.3 Dative alternation in the English of England – the Survey of English Dialects (left, top legend) and ‘give it me’ in the English Dialects App (right, bottom legend).

Understandably, the media needed to be central to our plans for dissemination of the app. We drew up lists of media organisations, newspapers, magazines, etc. to contact and began to sound some of them out, particularly the BBC because of their extensive network of local TV and radio stations. Having been impressed at how the Great British Class Survey (Savage et al. 2013) had benefited from BBC website involvement, we considered, and discussed with relevant BBC editors, the possibilities of co-hosting the survey on the Education and Science pages of their website. Making contact was extremely difficult, virtually impossible via the BBC website, and we had to rely on personal contacts to eventually ‘break in’. Despite long and serious discussions, this potential opportunity did not in the end come to fruition. We ended up following the rather familiar route of a press release via our university press offices. A rather long press release was written and was issued in mid-January 2016, both in England, from the University of Cambridge, where Adrian was based at the time, and in Switzerland, from the University of Bern, where Dave was working.

Impact 2: take off and relaunch

That Adrian was based at Cambridge at the time of the launch and that the release came from the university’s press office were factors that undoubtedly helped in piquing the media’s attention at what we were doing, and framing the project as a serious endeavour. The story was picked up by most national newspapers, and many TV and radio organisations, especially the BBC, and was widely forwarded and published both in printed and in online-only media formats, in Britain and overseas. Despite the ‘foreignness’ of the theme, Swiss media also picked up on the story, pointing to the fact that the app had been developed from an originally Swiss German one. In the first week, we conducted over 25 interviews about the app on British TV, national and local radio and with newspaper journalists. The app was downloaded over 38,000 times on the day of the launch and for a couple of days it was the most downloaded education-related app in the UK. By May 2016, over 43,000 people had completed the quiz and left their social metadata and 3,500 had uploaded recordings. We had gathered sufficient data by this stage to (i) improve the dialect predictability function of the app by basing it on the data we had already collected in the first five months rather than on the Survey of English Dialects data as well as (ii) release some maps showing the distribution of dialect variants based on quiz answers (such as Figure 7.3). Both Cambridge and Bern issued a new press release containing some of the first results in late May 2016. There was a small surge in app downloads following the release – 4,000 on 27 May – and the media attention that ensued. More interviews were conducted, there was widespread TV, radio and newspaper coverage, and even a mention on the BBC

satirical TV panel show *Have I Got News For You*. Following this peak in downloads, more data were collected.

Since the May 2016 press release, we have not actively sought any further media attention. What we have discovered, however, is that (i) the story continues to be picked up and (ii) now that the story and some of the results are in the public domain, they are being spread without citing the research or the researchers. Perhaps unsurprisingly, the variable that has received the most subsequent attention is the pronunciation of 'scone'. YouGov, the polling agency, conducted another survey in October 2016 on scones, including their pronunciation. The *Telegraph* in its online site (<http://www.telegraph.co.uk/food-and-drink/news/survey-reveals-correct-way-to-pronounce-scone/>) reported the results of the YouGov survey, but used our map, without attribution, and without mentioning that it didn't come from the YouGov survey. In the late autumn, it was submitted without attribution to Reddit's Mapporn site (and the submitter even attributed the map to himself ...), and online media started circulating our map under the title 'The Great Scone Map of the United Kingdom and Ireland'. We had not given it that title. Often its use was not credited to us. We felt that with our limited resources, the chances of a successful protest against the dissemination of our work without our permission were probably futile. It was widely posted on a number of online media platforms, and even baking companies picked it up: e.g. <https://www.haywood-padgett.co.uk/we-all-love-them-but-how-do-you-pronounce-it/>. In April 2017 it was re-reported (and this time correctly attributed) by the *Observer* and run by several other news organisations as a result, which led to a second mention on *Have I Got News For You*.

Impact 3: back down to earth – reflections on 'impact'

Experiences with the media

We came to the project with different experiences of the media, both old and new. Dave had had brief but frequent experiences of media interest in language and dialect issues, dating back to his PhD days in the late 1980s when a newspaper report had labelled him 'a dialect fanatic' after what had seemed a perfectly reasonable chat with a journalist. All media excursions since had involved very brief interviews and short reports, on themes such as 'Estuary English' and other dialects of the South-East of England, High Rising Terminals/'Uptalk' and the role of the media in language change, with one TV appearance discussing minority languages of the British Isles with a puppet ferret and Pop Idol singer Darius Campbell on Children's BBC. Apart from the latter, much of it felt like the function was to fill an otherwise empty column inch in a newspaper, or the 90 seconds before the radio news. Marie-José and Adrian, meanwhile, had had much more, and more recent experience, especially discussing the Dialäkt Äpp with Swiss media in a culture where the local Swiss German dialect is used by everyone and is culturally very prominent and prestigious, and at a time when social media already constituted a significant site for the circulation of news. This experience included interviews with TV and radio stations as well as with newspapers, where the researchers were even occasionally asked to write a draft of the article for the busy journalist. Once the data collected by Dialäkt Äpp had been analysed, several media outlets published Adrian and Marie-José's maps and graphics on dialect variation and change. Other work on Swiss German has led Adrian to field questions on what is the most popular dialect (of German) in Switzerland, which one is the most prestigious, which one is most likely to get you a job – what he calls the dialect hit parade. Dialect is taken very seriously in Switzerland, and so, for the most part, is the reporting of research on it. There continue to be loud debates, for example, on whether early education in Swiss German schools should be conducted in the local dialect, rather than in Standard German. Adrian and Marie-José also collaborated with the media in creating a dialect-guessing app for German-speaking Europe, which was also hosted on the media outlet's platform and gathered thereby an especially high number of users.

Our engagement with the media on the English Dialects App reflected to a large extent some of the advantages and cautions highlighted by Lawson (2016). We certainly benefitted in terms of sheer publicity from the BBC's organisational prowess and extensive network of local media stations. At one point, very shortly after the launch of the app, the BBC's General News Service (GNS), a corporation-internal news distribution centre, lined up a long list of local radio stations who wished to interview Dave, and those who were late for their GNS-allocated slot simply missed their chance. Hitting the main daily newspapers had a considerable knock-on effect too, as stories from these papers were picked up and re-transmitted or rewritten by other media outlets, online and not. Our 'story' suffered some but not extensive media sensationalisation, a commonly reported problem for sociolinguists trying to disseminate their research (Lawson 2016: 177). When we reported that we had evidence of several local dialect features gradually dying

out, and evidence of some variants associated with London spreading beyond the South-East, it was reported that we had said that 'English' was being 'killed off by the London dialect'. We were actually surprised how rarely this occurred though. Generally, our interactions with the media were friendly, respectful, engaging (and live, preventing them in the process from spinning some other 'telling' of the story). Except in the immediate aftermath of the app's two launches, however, media reports became vaguer about who had conducted the research. So whilst at the start it was reported as a collaboration between researchers at Cambridge, Bern and Zurich, and we reiterated this whenever we could in media interviews, over time, such as in the case of the 'Great Scone Map of the UK and Ireland', the researchers were labelled as 'Cambridge academics', even though, by that stage, Adrian had moved from Cambridge to Lancaster, or, in some of the online sources, incorrectly sourced or not accredited at all. If we'd needed hard evidence of some of these appropriations of our work as 'impact', we would have needed to accompany them with an explanatory narrative about why we and our institutions weren't named.

Our active social media engagement was limited to Facebook and Twitter, and we did not use these perhaps as vigorously as we might have. We were conscious that the app was going to reach digital natives relatively easily, and Twitter was probably not going to reach audiences we had not already accessed successfully through our liaising with media outlets. Also, after having shared information on previous dialect-predicting apps earlier, our Facebook followers' attention was possibly not as high as with Dialäkt Äpp. What was especially striking with the online content that emerged, especially in the aftermath of the second press release with its maps of results, was the degree of metalinguistic commentary that it generated in the online comments sections. The *Observer* article – online on the *Guardian* website – on the pronunciation of 'scone', for example, generated over 2,600 comments. Had we 'changed behaviour', to draw from the REF definition of impact? This is always hard to judge. Our work on the pronunciation of scone produced, for the first time, a 'geography' of the pronunciation of this so salient word, and this was raised by some in the online comments. It demonstrated (as is the want of sociolinguists ...) that variation is not free and random, even for a word demonstrating such robust variability as this one. Some commenters pushed home the point that there is no 'right' or 'wrong' way to pronounce words like this.

Probably because of relatively easy inter-site story sharing, news of the app and its results appeared on many online platforms we had previously been unaware of. A downside of this was, as mentioned earlier, the complete detachment of the results from the research and the researchers.

Beyond the media

Ultimately, we were relatively successful in our efforts to secure enough impact for the app to gather sufficient data for our dialectological needs – the app, as we expected, reached a rather skewed demographic – too many digital natives – but because we sought social metadata from each user, the app nevertheless enables us to control for social background, age, gender, mobility and so on. That we were able to relatively quickly produce colourful maps of results also aided this effort – data were collected in a way that they could be quickly and readily processed and charted. The inclusion of the scone question was a game-changer in many senses, as many media outlets led their stories with the 'scone' story after both the first launch (e.g. *The Telegraph*, BBC Cambridgeshire, *Yorkshire Post*) and the second (e.g. *The Guardian*, *The Times*), and re-found it as it was posted freely from then on around the internet. We needed this sort of coverage. We had to grit our teeth very occasionally at the way it was reported, but it worked well enough.

But can we say it created impact in a way that would make HEFCE and REF submitters happy? It would be hard to argue that it had had 'an effect on, change or benefit to the economy, society, culture, public policy or services, health, the environment or quality of life, beyond academia' (HEFCE 2011: 26). Easier to demonstrate is that 'public debate has been stimulated or informed by research' (HEFCE 2012: 28, see also Lawson and Sayers 2016: 14–15). This broadening of the scope of impact has been supported by the Stern (2016: 23) review of the REF, which recommended that 'impact case studies should not be narrowly interpreted, need not solely focus on socioeconomic impacts but should also include impact on ... public engagement and understanding, on cultural life, on academic impacts outside the field, and impacts on teaching'.

It also recognised that 'realising the full impact of research is a continuing and sometimes long-lived process' (Stern 2016: 23). Our app data, we would argue, does have the potential in the longer term to make 'impact' in the sense of HEFCE's original narrower definition – in the field of forensic phonetics. It is not a

potential that we have used in our media discussions about the app, and is one which will take much longer to come to fruition. We present it briefly here now.

The data collected with the English Dialects App has potential applications in forensic work on speaker profiling and speaker comparison. In speaker profiling, a recording of an unknown speaker is analysed acoustically and auditorily to derive as much information as possible from that speaker. Provided the audio recording is long enough, this may include a speaker's social and educational background, first language influences (e.g. a non-native accent), evidence of speech impediments, signs of intoxication, age (with cautions!), gender and regional background (see Jessen 2008). As a hypothetical example, the perpetrators of a masked robbery are captured on CCTV, their faces not seen, but their voices recorded. The purpose of speaker profiling is to assist the police by identifying or narrowing down sub-sections of wider populations to which a speaker belongs – a thorough acoustic and auditory analysis of the robbers might strongly suggest that the robbers are male, young and from the South-West of England. In forensic speaker comparison, on the other hand, a speech recording of an unknown speaker associated with a crime and a recording of someone suspected to be identical with the unknown speaker exist, let's say a kidnapper having been recorded while making a ransom demand. Here, recordings are compared with respect to several features, e.g. fundamental frequency, speaking rate, formant frequencies, rhythm, voice quality, etc. Forensic phoneticians then reach a conclusion how likely it is that the samples come from the same speaker. A written report is drafted and, depending on the legal system, may be used as evidence in court.

In both cases – speaker profiling and speaker comparison – the speakers' regional accents play a central role. Findings of a survey on forensic speaker comparison practices found that 28% of practitioners examine dialect and accent features (Gold and French 2011). For speaker profiling, too, Köster et al. (2012) found that regional accents are one of the most powerful tools to identify a speaker. They found that the forensic teams use different strategies to identify a speaker's regional background: they may perform auditory and acoustic analyses and/or consult audio databases and literature on dialects. The latter category includes the consultation of dialect Atlases. As stated earlier, much of the literature on geographical accent and dialect variation in England is both outdated and based on a very restricted and non-mobile demographic. Our new maps are able to provide an updated picture of regional variation for many features across England, providing forensic experts with a comprehensive database of unprecedented spatial resolution.

The app's corpus of audio recordings, too, has relevance for speaker comparison. When experts compare disputed samples (recordings of the perpetrator) and known samples (recordings of the suspect), they look for a high similarity in the two speech signals, e.g. high similarity in speech rate, rhythm or sound formant frequencies. At the same time, the features compared should ideally have low typicality (i.e. not occur very frequently) in the population, as this increases the likelihood of the samples coming from the same speakers (French and Harrison 2007; Jessen 2008). One example is where known (the police know, for example, that the sample comes from the accused) and disputed (the police believe it comes from the accused, but this is disputed by the accused) speech samples both show very fast articulation rates – such very fast articulation rates occur very infrequently in the population as a whole (see Parkinson 2017), i.e. these manifestations have what is known as low typicality and would suggest an increased likelihood that the two samples are in fact from the same speaker. If, on the other hand, the known and disputed samples both show average speeds of articulation – average articulation rates are very frequent, i.e. they have high typicality – this decreases the likelihood that the two samples come from the same speaker. Forensic phoneticians, however, need to firmly establish if a particular feature, such as rhythm, voice quality, speech rate, has low or high typicality, and to do this they need to know how such features are distributed across the population as a whole, in different places, across different age groups and among people with different social backgrounds. Apart from a few global speech characteristics (such as f_0 , the fundamental frequency of the speech wave, for example; see Hudson et al. 2007), there is a general lack of information about how speech features are distributed across the population because of the lack of large scale controlled corpora of readily analysable spoken data. A small pilot study, however, shows promise for the potential of the app in this domain. Hudson et al. (2016) analysed fundamental frequency (f_0) statistics from 2,000 of the app's uploaded recordings. Male adults in the corpus have a mean f_0 of 129 Hz. Reassuringly, the mean f_0 in the app data is almost identical to that reported by Johns-Lewis (1986) who finds a 128 Hz average for British English reading. The app's nationwide sample of over 4,300 recordings could provide a first account of how features such as speaking rate, voice quality, speech rhythm etc. are currently distributed in the UK speaker population as a whole, but also regionally and socially. Such analysis, and consequently the potential long-term impact of this aspect of the

app, takes much longer, however.

The 'impact agenda'

The penetration of the New Public Management paradigm of state governance which Lawson and Sayers (2016) argue was ultimately responsible for the impact agenda in British universities has been slower to arrive in Switzerland than in the UK, and, in comparison, its effects on universities and their research have been relatively light touch, so far. Although *Qualitätssicherung und -Entwicklung* (QSE) (Quality assurance and development) is an increasingly important theme within Swiss universities, and there are periodic evaluation-based surveys of its effectiveness in order for universities to secure accreditation, there is no equivalent of the UK's Research Excellence Framework or the earlier Research Assessment Exercise. Indeed, there is no national system at all for the evaluation of the research of individuals or their academic units, and individual universities and/or faculties within universities devise their own ways of collating and tracking research output. This can perhaps be explained by Switzerland's federal system, in which each canton has a great deal of sovereignty – universities are cantonal, not national, institutions. The *Hochschulförderungs- und -koordinationsgesetz* (HFKG) (Higher Education Funding and Coordination Act) simply requires that research be included within QSE as part of the accreditation process. Universities essentially operate their own methods of doing this. The University of Bern's website states that the university 'sieht den Nutzen der Evaluationen in der Möglichkeit der Profilierung der Einheiten, dieser soll mit einem möglichst geringen Aufwand für die Forschenden erzielt werden' – it sees the opportunity to profile its research units as the *point* of the exercise, with *minimum effort on the part of the researchers*.² In Dave's faculty in Bern, for example, staff are encouraged to upload details of publications, presentations and other outputs onto an online repository, and these form part of an annual departmental report, alongside other activities such as administrative service, the organisation of conferences, peer review activities, media activity, prizes and other esteem indicators. Similar conventions apply at the University of Zurich.

Naturally, research impact in a general sense is seen as a Good Thing within the system, and an increase in the 'Sichtbarmachung' (making visible) of research, especially in the humanities, is stated as an important strategic goal of *swissuniversities*, the collaborative body of Swiss institutions of higher education. Both universities, largely reliant on their local cantons for funding (with very little student fee income) and the national science foundation, the SNF, engage actively and visibly with the non-academic community and organize, for example, regular town and gown-type events, as well as publish glossy research communication magazines, such as *Horizonte*. The SNF funds science communication under the AGORA scheme that 'aims to foster dialogue between scientists and society'. After the success of the Dialäkt Äpp, Adrian, Marie-José and their colleagues obtained such funding and developed the Voice Äpp, which predicts users' dialect based on automatic speech recognition. When applying for regular research grants from the SNF, applicants must write a lay summary of their project, and, if successful, upload information to a central repository not only about their academic research and publications but also about 'knowledge transfer events',³ 'public communication' and 'use-inspired outputs'. But the *requirement* to demonstrate 'impact' in the sense of the REF and the evaluation of impact has not yet arrived.

Coming from the UK's highly regulated system of surveillance and quality assurance and the RAE system (he left before the REF) to the Swiss system, Dave was rather shocked, not by the lack of an RAE, but by the extent to which he had internalised some of its assumptions and premises – the lack of urgency to publish in the major international peer-reviewed journals, the preference for monographs over journals as a research outlet, the way that, in the assessment of annual reports a journal article counted as a journal article, no matter which journal it was published in, a book was a book, no matter with which publisher. Refreshing indeed was the bizarre idea that a professional academic could set up a course, teach and assess it without the need for multilayered teams of non-experts in that subject poring over every detail, demanding greater evidence of the integration of key skills or tut-tutting at aspects of the assessment detail. Adrian went from Switzerland to the UK, first to Cambridge, then Lancaster. There he faced a much greater, explicit day-to-day orientation to these audit mechanisms, to the Teaching Excellence Framework (TEF), the Research Excellence Framework (REF), to league tables, to his own university's performance relative to that of others and to active policy change that might boost the department or university's position in the latest metric. Having, in a sense, swapped places, we often share stories of our amazement at how our new academic cultures function, and see good and less good in both of these very different systems. Of relevance here is that while

both systems celebrate impact, only one of them appears to require it, measure it, compare it, rank it and fund on the basis of it.

Marie-José, meanwhile, went from Swiss academia to science and data-driven publishing with NZZ, Switzerland's leading quality newspaper. She mainly works on longer-term projects, sometimes also on short-term articles that are heavily news-related. The working process is often similar to that of her former academic work: finding an interesting question or hypothesis to investigate, gathering data, processing, analysing and visualising data, interpreting it and writing it up. When having to deal with the media as an academic, she was often surprised by how fast journalists needed an interviewee to answer their questions. If they could not reach a specific expert within a few minutes, they would just call the next one whose number they could find. Also, it was sometimes frustrating how journalists seemed to 'water down' scientists' statements or present them out of context or in a way that was not sufficiently nuanced. As a journalist, however, she has experienced how fast some topics and therefore articles need to go online (old news simply has less impact), how little time there is to wait for a specific researcher to pick up the phone – someone else from this or another lab will often be just as qualified to give answers on the topic. And she has appreciated the importance of wording and explaining complex matters in a way that is not only accurate, but also understandable to a lay readership. Scientists read papers out of intrinsic interest for the topic, and are ready and prepared to be challenged with long, complex sentences, abstract lexicon and very specialised terminology. Newspaper readers, however, will take an interest in many topics, but only if the story is told in an engaging way.

At a recent workshop with PhD students and postdoctoral staff of the University of Zurich, Marie-José was invited to talk about how to reach out to the media, based on her experience (i) with Dialäkt Äpp, and (ii) as a data journalist. Many of the participants had little or no experience with the media, but were interested in outreach – not least because they felt they wanted to give something back to the tax payers who funded them, and to disrupt the 'filter bubble' between perceived elites and the general public. Others felt they were mainly doing a service to their university by communicating to the media, as the university is funded by the public and nudges them to reach out to this public. However, there was a general feeling that a lack of outreach experience on a CV is not problematic when applying for senior academic positions, at least in Switzerland. When they apply for grants, they tend to stress their willingness to communicate the results of their research to a wider public. Some reported that the scientific advisory board of a project they were involved in specifically suggested that the project should reach out to the media more. And many reported that, when choosing a research subject, it would be helpful if it were understandable to a broader public, as fundamental research was deemed difficult to communicate.

Conclusion

For many of us, stepping outside of our normal university routines of teaching, administration and research is not easy. We are understandably much more used to talking to each other about what we do than to people outside. We are sure that journalists having to engage with academics are as perplexed and frustrated with us as we not infrequently are with them. In many cases we are not given the time to pursue this kind of dissemination of our research, or we juggle it with a myriad of other expectations. In the case of the English Dialects App, the success of our more academic interests depended on the success of this media engagement, however. We had experience in the team – both Marie-José and Adrian had considerable experience of the media attention given to mass crowdsourcing projects. We had the help of the Press Office at Cambridge University, who helped us shape our message in a way that would hopefully appeal, but then we were on our own. We secured enough coverage to be able to harvest a sufficient amount of data for our purposes. The immediate impact of the app was probably initially to entertain for a few minutes. We are sure that, out there, a good number of people must have had a laugh at the app's dialect prediction for them. We received many emails from users telling us long stories of their own mobility histories and how the app had 'located' them to one of the places on their route. With the release of the first results from the app, we were able, hopefully, to ensure that public debate had been stimulated and informed about dialect variation and language change. We hope the maps created by the app will, for example, eventually serve as useful materials for school English classes, to exemplify geographical variation and the spatial distribution of linguistic change. If we are to make any significant *long-term* non-academic impact at all, however, it may well be as a result of the slower, rather more technical exploitation of the app's large dataset, for example in

speech recognition, speech technology more generally, and in forensics. We may, then, in years to come, be able to argue that, indirectly, scones have played a small role in the criminal justice system.

Notes

- 1 We are absolutely indebted to the following people for their assistance and collaboration in the building, dissemination and analysis of the app and its results: Daniel Wanitsch, Tam Blaxter, Sarah Grossenbacher, Melanie Calame and Heidi Suter.
- 2 http://www.unibe.ch/forschung/gute_wissenschaftliche_praxis/qualitaet_in_der_forschung/forschungsevaluation/index_ger.htm
- 3 'Wissenstransfer' is used in the official German text and not 'Wissensaustausch' – knowledge exchange. Grant holders are told they should upload information about 'transferring knowledge to non-scientific experts'.